

Q15

The p.d. across the 4Ω resistor, $V = IR = 1.5 \times 4 = 6.0\text{ V}$

Hence the current in the 13Ω resistor, $I = \frac{V}{R} = \frac{6.0}{13} = 0.4615\text{ A}$

The total current flowing in the internal resistor = $0.4615 + 1.5 = 1.9615\text{ A}$

Using $P = IV$, we get $V = \frac{2.1}{1.9615} = 1.0706 = 1.1 \text{ V}$

Hence emf = total pd = $3.0 + 6.0 + 1.1 = 10.1 \text{ V}$

Ans: D